



Part 1: Fundamentals & Classical Statistical Methods

What is a Time Series?

A **time series** is a sequence of observations collected **over time**, usually at regular intervals.

Examples:

- Daily stock prices
- Hourly electricity consumption
- Monthly sales revenue
- Sensor measurements every second

Why Time Series Are Different

- Observations are **dependent**
- The order of data points **cannot be shuffled**
- Past values influence future values

This dependency structure is the core challenge of time series analysis.

Time Series Notation & Properties

Indexing Observations with t

In time series, each observation is indexed by a time step t :

Notation	Meaning
Y_t	Current observation (at time t)
Y_{t-1}	One period before t — also called lag 1
Y_{t-2}	Two periods before t — lag 2
Y_{t+1}	One period ahead of t (used when forecasting)

Example — if today is Wednesday and you have daily data:

- Y_t = Wednesday's value
- Y_{t-1} = Tuesday's value
- Y_{t-2} = Monday's value

This notation is used throughout time series models. For instance, an AR(1) model is written as:

$$Y_t = c + \varphi \cdot Y_{t-1} + \varepsilon_t$$

meaning: "today's value depends linearly on yesterday's value plus random noise."

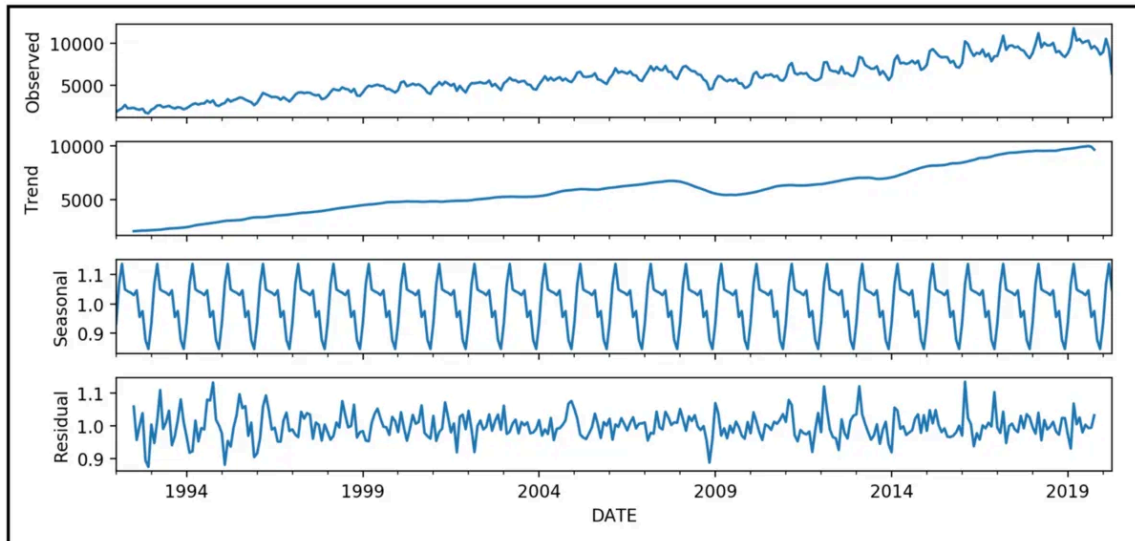
Important Properties of Time Series

- **Regular (constant) time intervals** — observations must be evenly spaced: daily, weekly, monthly, etc. Missing timestamps should be explicitly represented (filled or marked as NaN), not silently skipped — gaps in the index change the meaning of every lag.
- **Temporal ordering matters** — the sequence cannot be shuffled. The past influences the future, so the order of rows is data, not decoration.
- **Lag** — the distance between the current time step and a past one. Lag k refers to k periods back: Y_{t-k} . Models use lags as features.
- **Stationarity** — many models assume the mean and variance of the series remain constant over time. Non-stationary series must usually be transformed (e.g. by differencing) before modelling.

Key Components

Time series typically contain four components:

1. **Trend (T)**: Long-term increase or decrease in the data
2. **Seasonality (S)**: Regular, predictable patterns that repeat over fixed periods (e.g., yearly, monthly)
3. **Cyclicity (C)**: Patterns that repeat but not at fixed intervals (e.g., economic cycles)
4. **Noise/Irregular (I)**: Random variation that cannot be attributed to trend, seasonality, or cyclicity



Mathematical Representation

- **Additive Model:** $Y(t) = T(t) + S(t) + C(t) + I(t)$
 - Use when seasonal variation is roughly constant over time
- **Multiplicative Model:** $Y(t) = T(t) \times S(t) \times C(t) \times I(t)$
 - Use when seasonal variation increases with the level of the series

Time Series Analysis in Python - Setup and Imports

Before we begin, install the required packages:

```
pip install pandas numpy matplotlib statsmodels scipy scikit-learn
```

```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.stattools import adfuller, kpss
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.stats.diagnostic import acorr_ljungbox
from scipy import stats
from sklearn.metrics import mean_absolute_error, mean_squared_error
import warnings
warnings.filterwarnings('ignore')

plt.style.use('seaborn-v0_8-darkgrid')
```

Creating a Sample Time Series

The code below makes fake daily data that looks like a real-world time series.

More specifically:

- It creates **daily dates from 2020 to 2023**.
- It makes the values **slowly increase over time**.
- It adds a **repeating yearly up-and-down pattern**.
- It adds a bit of **random randomness** so it is not perfectly smooth.
- It combines all of that into one dataset and shows a small sample.

In short:

It simulates realistic daily data with a trend, seasonality, and noise.

```
In [21]: # Set random seed for reproducibility
np.random.seed(42)

# Generate date range
date_range = pd.date_range(start='2020-01-01', end='2023-12-31', freq='D')
n = len(date_range)

# Components
trend = np.linspace(100, 150, n) # Linear trend from 100 to 150
seasonality = 10 * np.sin(2 * np.pi * np.arange(n) / 365.25) # Yearly seasonality
noise = np.random.normal(0, 5, n) # Random noise

# Combine components (additive model)
ts_data = trend + seasonality + noise

# Create pandas Series with datetime index
ts = pd.Series(ts_data, index=date_range, name='Value')

print("Sample Time Series:")
print(ts.head(10))
print(f"\nShape: {ts.shape}")
print(f"Period: {ts.index.min()} to {ts.index.max()}")
```

Sample Time Series:

```
2020-01-01    102.483571
2020-01-02     99.514941
2020-01-03    103.650916
2020-01-04    108.233733
2020-01-05     99.653774
2020-01-06     99.859609
2020-01-07    109.131857
2020-01-08    105.278161
2020-01-09     99.298455
2020-01-10    104.563060
```

Freq: D, Name: Value, dtype: float64

Shape: (1461,)

Period: 2020-01-01 00:00:00 to 2023-12-31 00:00:00

Visualizing the Time Series

The code below draws a line chart of the time-series data.

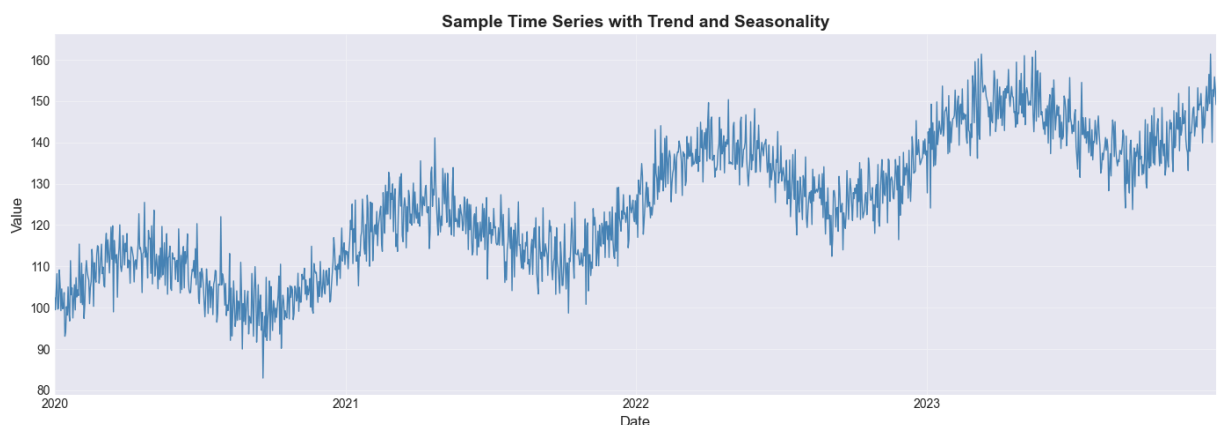
More specifically:

- It creates a **plotting area** of a fixed size.
- It **plots the time series as a line** on that area.
- It adds a **title** and **labels** for the x- and y-axes.
- It turns on a **light grid** to make the chart easier to read.
- It adjusts spacing so labels are not cut off.
- It **displays the chart**.

In short:

It visualizes the time series so you can see the trend and seasonal pattern over time.

```
In [22]: #subplot function returns a tuple (figure, axes), figure is the entire figure
# Axes object is what we plot on
# figsize is in inches
fig, ax = plt.subplots(figsize=(14, 5))
ts.plot(ax=ax, linewidth=1, color='steelblue')
ax.set_title('Sample Time Series with Trend and Seasonality', fontsize=14, f
ax.set_xlabel('Date', fontsize=12)
ax.set_ylabel('Value', fontsize=12)
ax.grid(True, alpha=0.3)
plt.tight_layout() # Adjust layout to prevent clipping of labels
plt.show()
```



2. Time Series Decomposition

Theory

Decomposition is the process of separating a time series into its constituent components. This helps us:

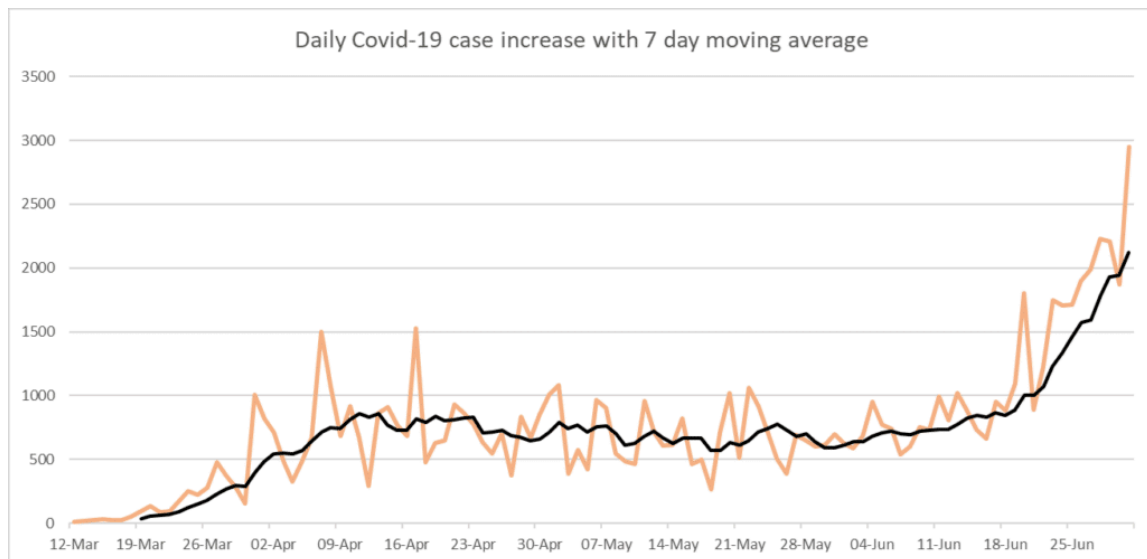
- Understand the underlying patterns
- Remove seasonality for better modeling
- Identify anomalies
- Choose appropriate forecasting methods

Classical Decomposition Methods

The most common method is **moving average decomposition**:

1. Estimate trend using moving average
2. Detrend the series
3. Estimate seasonal component by averaging detrended values for each season
4. Calculate residuals: $\text{Residual} = \text{Original} - \text{Trend} - \text{Seasonal}$

Month	Sales (\$000)	Three-month moving total (\$000)	Three-month moving average (\$000)	Seasonal variation (\$000)
January	125			
February	145	456=(125+145+186)	$(456 \div 3) = 152$	
March	186	462=(145+186+131)	$(462 \div 3) = 154$	
April	131	468=(186+131+151)	$(468 \div 3) = 156$	
May	151	474	158	
June	192	480	160	
July	137	486	162	
August	157	492	164	
September	198	498	166	
October	143	504	168	
November	163	510	170	
December	204			



Performing Decomposition

The code below breaks the time series into its main parts.

More specifically:

- It tells Python the data has a **yearly pattern** (365 days).
- It **separates the data** into:
 - a **trend** (long-term direction),
 - a **seasonal pattern** (repeating yearly ups and downs),
 - and **random noise** (what's left over).
- It stores each part separately.
- It prints the size of each component to confirm they match the original data.

In short:

It decomposes the time series into trend, seasonality, and noise.

```
In [23]: # Perform additive decomposition
# Period = 365 because we have daily data with yearly seasonality
decomposition = seasonal_decompose(ts, model='additive', period=365)

# Extract components
trend_component = decomposition.trend
seasonal_component = decomposition.seasonal
residual_component = decomposition.resid

print("Decomposition Components:")
print(f"Trend:      {trend_component.shape}")
```

```
print(f"Seasonal: {seasonal_component.shape}")
print(f"Residual: {residual_component.shape}")
```

Decomposition Components:

Trend: (1461,)
 Seasonal: (1461,)
 Residual: (1461,)

Visualizing Decomposition

The code below draws four stacked charts to show the decomposition results.

More specifically:

- It creates **four plots arranged vertically**.
- The first plot shows the **original time series**.
- The second shows the **trend** (long-term movement).
- The third shows the **seasonal pattern** (repeating yearly cycle).
- The fourth shows the **residuals** (random noise left over).
- It labels each plot so you can clearly see what each line represents.
- It draws a zero line on the residual plot to make deviations easy to spot.
- Finally, it displays the figure.

In short:

It visually explains how the original data is split into trend, seasonality, and noise.

```
In [24]: fig, axes = plt.subplots(4, 1, figsize=(14, 10))

# Original
ts.plot(ax=axes[0], linewidth=1, color='black')
axes[0].set_ylabel('Original', fontsize=11, fontweight='bold')
axes[0].set_title('Time Series Decomposition (Additive Model)', fontsize=14,

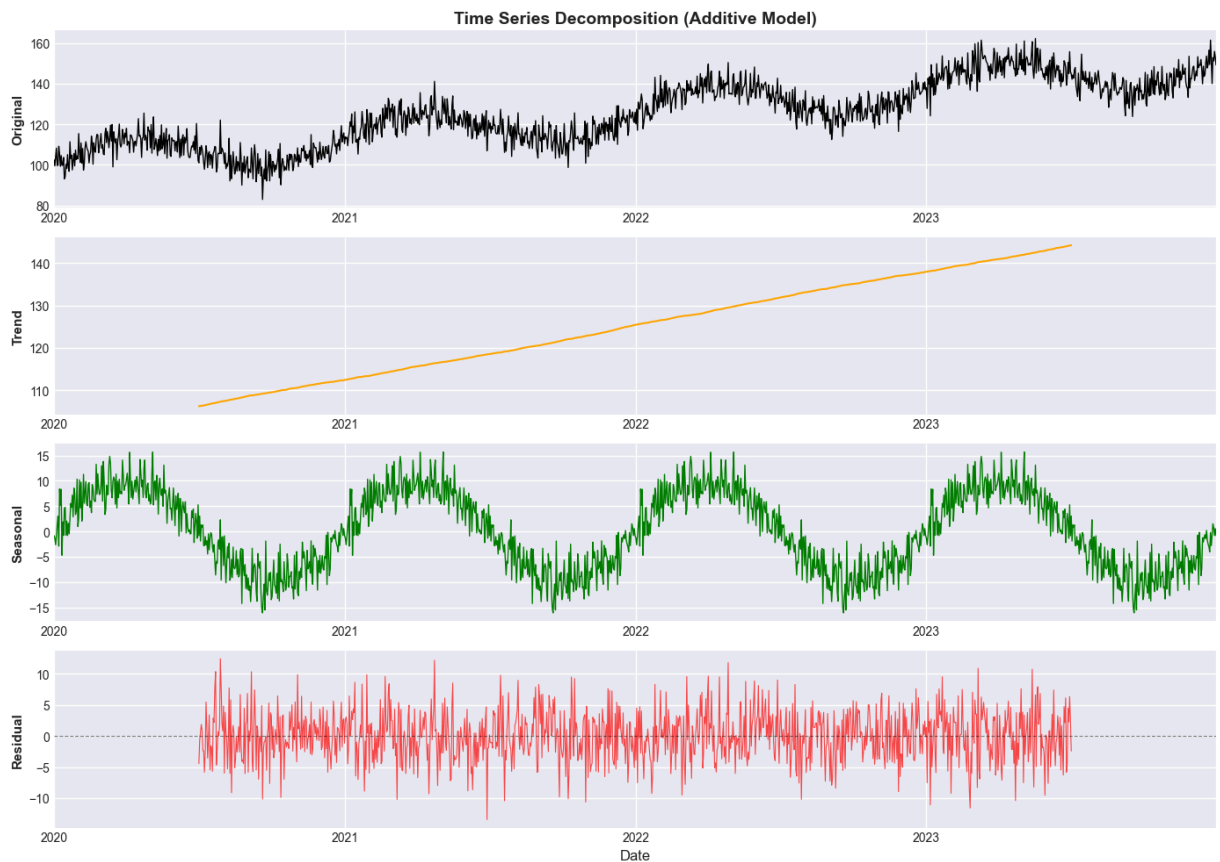
# Trend
trend_component.plot(ax=axes[1], linewidth=1.5, color='orange')
axes[1].set_ylabel('Trend', fontsize=11, fontweight='bold')

# Seasonal
seasonal_component.plot(ax=axes[2], linewidth=1, color='green')
axes[2].set_ylabel('Seasonal', fontsize=11, fontweight='bold')

# Residual
residual_component.plot(ax=axes[3], linewidth=0.8, color='red', alpha=0.7)
axes[3].set_ylabel('Residual', fontsize=11, fontweight='bold')
axes[3].set_xlabel('Date', fontsize=12)
axes[3].axhline(y=0, color='black', linestyle='--', linewidth=0.8, alpha=0.5)
```



```
plt.tight_layout()
plt.show()
```



3. Stationarity

What is Stationarity?

A time series is **stationary** if its statistical properties (mean, variance, autocorrelation) do not change over time.

Why Does Stationarity Matter?

Most classical time series models (ARIMA, SARIMA) assume stationarity because:

- Statistical properties are easier to model when constant
- Predictions are more reliable
- Mathematical theory is simpler

Types of Stationarity

1. **Strict Stationarity:** Joint distribution is time-invariant (very restrictive)
2. **Weak/Covariance Stationarity:** Only mean, variance, and autocorrelation are constant (commonly used)

Common Non-Stationary Patterns

- **Trend:** Mean changes over time
- **Seasonality:** Pattern repeats at regular intervals
- **Heteroscedasticity:** Variance changes over time

Augmented Dickey-Fuller (ADF) Test

The ADF test checks the null hypothesis that the series has a unit root (non-stationary).

- H_0 : Series has a unit root (non-stationary)
- H_1 : Series is stationary

Decision Rule: If p-value < 0.05, reject H_0 (series is stationary)

```
In [25]: def check_stationarity(timeseries, name='Series'):
    """
    Perform Augmented Dickey-Fuller test
    """
    print(f"\n{'='*60}")
    print(f"Stationarity Test: {name}")
    print(f'{'='*60}')

    # Remove NaN values
    ts_clean = timeseries.dropna()

    # Perform ADF test
    result = adfuller(ts_clean, autolag='AIC')

    print(f'ADF Statistic:      {result[0]:.6f}')
    print(f'p-value:              {result[1]:.6f}')
    print(f'# Lags Used:            {result[2]}')
    print(f'# Observations:         {result[3]}')
    print(f'\nCritical Values:')
    for key, value in result[4].items():
        print(f'  {key}: {value:.3f}')

    # Interpretation
    print(f'\n' + '-'*60)
    if result[1] <= 0.05:
        print(f"✓ STATIONARY (p={result[1]:.4f} < 0.05)")
        print("  → Reject null hypothesis")
    else:
        print(f"✗ NON-STATIONARY (p={result[1]:.4f} > 0.05)")
        print("  → Fail to reject null hypothesis")
    print(f'{'-'*60}')

    return result[1] <= 0.05
```

Testing Our Series

```
In [26]: # Test original series
is_stationary = check_stationarity(ts, 'Original Time Series')
```

```
=====
Stationarity Test: Original Time Series
=====
```

```
ADF Statistic:      -0.835398
p-value:           0.808502
# Lags Used:       12
# Observations:    1448
```

```
Critical Values:
```

```
1%: -3.435
5%: -2.864
10%: -2.568
```

```
-----
x NON-STATIONARY (p=0.8085 > 0.05)
  → Fail to reject null hypothesis
-----
```

Making a Series Stationary using the Differencing Method

Differencing removes trends and can stabilize the mean:

1. First Difference

Original series: $Y(t)$

First difference measures **change between consecutive periods**:

$$Y'(t) = Y(t) - Y(t - 1)$$

Example:

- ($Y = [2, 5, 8, 11]$)
- First difference:

$$Y' = [5 - 2, 8 - 5, 11 - 8] = [3, 3, 3]$$

This removes a **linear trend**.

2. Second Difference (what it really means)

The **second difference** is simply the **difference of the first difference**.

$$Y''(t) = Y'(t) - Y'(t - 1)$$

In words:

“How is the change itself changing?”

3. Numerical Example

Start with a series that has **curvature** (not just a straight line):

Original data:

$$Y = [1, 4, 9, 16]$$

Step 1: First difference

$$Y' = [4 - 1, 9 - 4, 16 - 9] = [3, 5, 7]$$

Notice the first differences are **increasing**, so the trend is not linear.

Step 2: Second difference

$$Y'' = [5 - 3, 7 - 5] = [2, 2]$$

Now the series is **constant** → the curvature has been removed.

4. When to use Second Differencing

- Use **first differencing** → remove linear trend
- Use **second differencing** → remove quadratic / accelerating trend
- In practice, ARIMA models almost always use **d = 0 or 1**
- **d = 2** is rare and should be used cautiously

```
In [27]: # Apply first differencing
         ts_diff = ts.diff().dropna()

         # Test differenced series
         is_stationary_diff = check_stationarity(ts_diff, 'First Differenced Series')
```

```
=====
Stationarity Test: First Differenced Series
=====
```

```
ADF Statistic:      -17.823692
p-value:            0.000000
# Lags Used:        11
# Observations:     1448
```

```
Critical Values:
```

```
1%: -3.435
5%: -2.864
10%: -2.568
```

```
-----
✓ STATIONARY (p=0.0000 < 0.05)
  → Reject null hypothesis
-----
```

Visualizing the Transformation

The code below compares the original data with a transformed version that removes trend.

More specifically:

- The top plot shows the **original time series**, which changes over time (non-stationary).
- The bottom plot shows the **first-differenced series**, where each value is replaced by the change from the previous one.
- The differenced series fluctuates around zero, making it **more stable (stationary)**.
- The horizontal zero line helps you see this stability.
- Both plots share the same time axis for easy comparison.

In short:

It shows how differencing turns a trending series into a stationary one.

```
In [28]: fig, axes = plt.subplots(2, 1, figsize=(14, 8), sharex=True)

# Original
ts.plot(ax=axes[0], linewidth=1.2, color='steelblue')
axes[0].set_title('Original Series (Non-Stationary)', fontsize=12, fontweight='bold')
axes[0].set_ylabel('Value', fontsize=11)
axes[0].grid(True, alpha=0.3)

# Differenced
ts_diff.plot(ax=axes[1], linewidth=1, color='orange')
axes[1].set_title('First Differenced Series (Stationary)', fontsize=12, fontweight='bold')
axes[1].set_ylabel('Differenced Value', fontsize=11)
axes[1].axhline(y=0, color='red', linestyle='--', linewidth=1, alpha=0.7)
axes[1].set_xlabel('Date', fontsize=12)
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



4. Autocorrelation Analysis

ACF: Autocorrelation Function

The ACF measures the correlation between observations at different time lags:

$$\text{ACF}(k) = \text{Corr}(Y_t, Y_{t-k})$$

- Lag 1: correlation between consecutive observations
- Lag 2: correlation between observations 2 time periods apart
- And so on...

PACF: Partial Autocorrelation Function

The PACF measures the correlation between Y_t and Y_{t-k} after removing the effect of intermediate lags.

Why Are ACF and PACF Important?

They help identify the order of AR and MA components:

Pattern	Model Suggestion
PACF cuts off after lag p , ACF decays	AR(p)

Pattern	Model Suggestion
ACF cuts off after lag q , PACF decays	MA(q)
Both decay gradually	ARMA(p, q)

Plotting ACF and PACF

The code below looks at how the data is related to its past values.

More specifically:

- It uses the **stationary (differenced) series**.
- The **ACF plot** shows how today's value is correlated with previous days.
- The **PACF plot** shows the *direct* effect of past days, removing indirect effects.
- The plots help decide how many past values a time-series model (like ARIMA) should use.

In short:

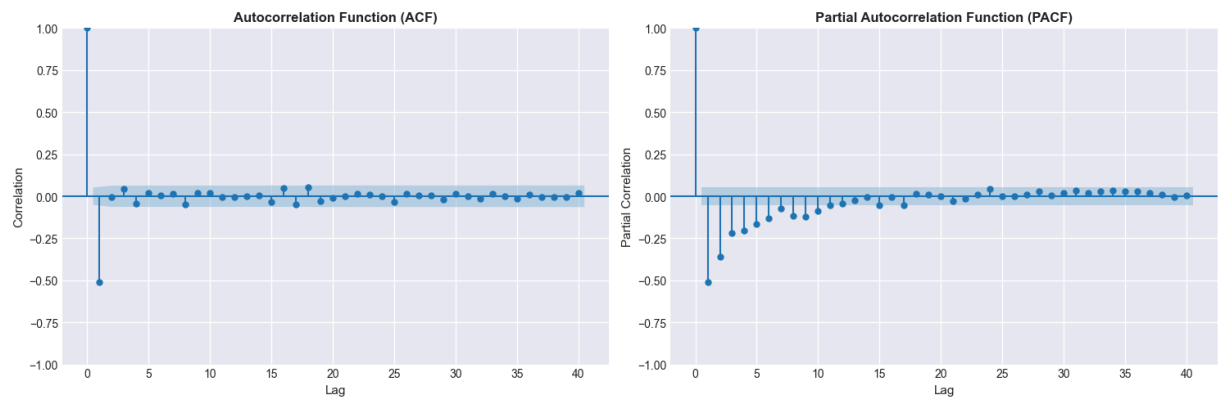
It helps you choose the right ARIMA model by showing which past values matter.

```
In [29]: # Use differenced series (should be stationary)
fig, axes = plt.subplots(1, 2, figsize=(15, 5))

# ACF
plot_acf(ts_diff, lags=40, ax=axes[0])
axes[0].set_title('Autocorrelation Function (ACF)', fontsize=12, fontweight=
axes[0].set_xlabel('Lag', fontsize=11)
axes[0].set_ylabel('Correlation', fontsize=11)

# PACF
plot_pacf(ts_diff, lags=40, ax=axes[1])
axes[1].set_title('Partial Autocorrelation Function (PACF)', fontsize=12, fo
axes[1].set_xlabel('Lag', fontsize=11)
axes[1].set_ylabel('Partial Correlation', fontsize=11)

plt.tight_layout()
plt.show()
```



Interpretation:

- Blue shaded area represents the confidence interval
- Bars outside this area are statistically significant
- Look for where bars drop inside the confidence interval (cutoff point)

The ACF plot shows a single strong and statistically significant spike at lag 1, followed by correlations that lie within the confidence bounds at higher lags. This indicates a clear cutoff in the autocorrelation function after the first lag.

The PACF plot does not exhibit a sharp cutoff. Instead, the partial autocorrelations decay gradually toward zero over several lags, indicating a tailing-off pattern.

This combination—ACF cutting off at lag 1 and PACF tailing off—is characteristic of a moving-average process of order 1. When applied to a first-differenced series, this pattern is consistent with an ARIMA(0,1,1) model.

5. Classical Time Series Models

Model	Full Name	Equation	When to Use
AR(p)	Autoregressive	$Y_t = c + \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \epsilon_t$	PACF cuts off
MA(q)	Moving Average	$Y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$	ACF cuts off
ARMA(p,q)	AR + MA	Combines both	Both ACF & PACF decay
ARIMA(p,d,q)	Integrated ARMA	ARMA on d-times differenced data	Non-stationary series
SARIMA(p,d,q) (P,D,Q,s)	Seasonal ARIMA	Adds seasonal components	Seasonal patterns

Preparing Data for Modeling

Let's create a simpler dataset to demonstrate the models:

The code below **creates a simple autoregressive time series and splits it into training and test data.**

More specifically:

- It generates **300 days of fake daily data.**
- Each day's value depends on **70% of the previous day's value**, plus some random noise. This is an **AR(1) process.**
- The result is a series where values are **correlated with their immediate past.**
- The data is then split into:
 - **80% for training** (used to fit a model),
 - **20% for testing** (used to evaluate predictions).
- Finally, it prints how many observations are in each part.

In short:

The code simulates a simple AR(1) time series and prepares it for model training and evaluation.

```
In [30]: # Generate AR(1) process
np.random.seed(123)
n = 300
dates = pd.date_range(start='2022-01-01', periods=n, freq='D')

# AR(1): Y(t) = 0.7 * Y(t-1) + noise
ar_coef = 0.7
ar_data = [0]
for i in range(1, n):
    ar_data.append(ar_coef * ar_data[i-1] + np.random.normal(0, 1))

ts_simple = pd.Series(ar_data, index=dates, name='Value')

# Train-test split (80-20)
train_size = int(len(ts_simple) * 0.8)
train = ts_simple[:train_size]
test = ts_simple[train_size:]

print(f"Train: {len(train)} observations")
print(f"Test: {len(test)} observations")
```

Train: 240 observations

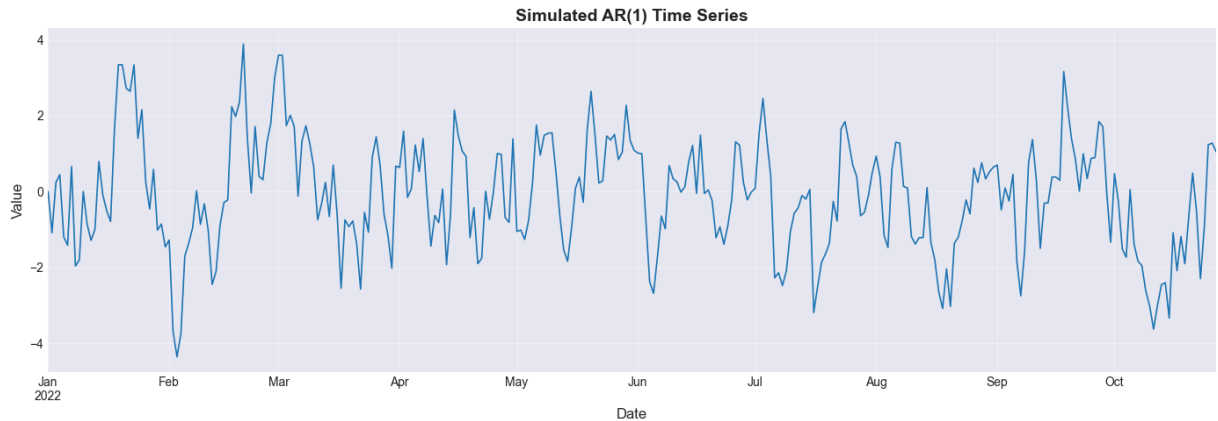
Test: 60 observations

Plot the dataset

```
In [31]: import matplotlib.pyplot as plt

# Plot the AR(1) time series
```

```
plt.figure(figsize=(14, 5))
ts_simple.plot(linewidth=1.2)
plt.title('Simulated AR(1) Time Series', fontsize=14, fontweight='bold')
plt.xlabel('Date', fontsize=12)
plt.ylabel('Value', fontsize=12)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Model 1: AR (Autoregressive)

Theory

An AR(p) model predicts the current value using p past values:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \epsilon_t$$

Where:

- c is a constant
- $\phi_1, \phi_2, \dots, \phi_p$ are coefficients
- ϵ_t is white noise error

Implementation

The code below **fits an AR(1) model to the training data and compares what the model learns to the true value used to generate the data.**

More specifically:

- It fits an **ARIMA(1,0,0)** model, which is an **AR(1)** model (one autoregressive term, no differencing, no moving average).
- The model is trained using the **training portion** of the simulated series.
- It prints a **summary** showing estimated parameters and diagnostics.
- It extracts the **estimated AR(1) coefficient** from the fitted model.

- It compares that estimate to the **true coefficient (0.7)** used to generate the data.

In short:

The code checks whether an AR(1) model can correctly recover the underlying dependency in the simulated time series.

```
In [32]: # Fit AR(1) model
# ARIMA(p,d,q) with p=1, d=0, q=0
ar_model = ARIMA(train, order=(1, 0, 0))
ar_fit = ar_model.fit()

print("AR(1) Model Summary:")
print(ar_fit.summary())
print(f"\nEstimated coefficient: {ar_fit.params[1]:.4f}")
print(f"True coefficient: {ar_coef}")
```

AR(1) Model Summary:

SARIMAX Results

```

=====
==
Dep. Variable:          Value    No. Observations:          2
40
Model:                ARIMA(1, 0, 0)    Log Likelihood          -345.7
82
Date:                Sun, 04 Jan 2026    AIC          697.5
63
Time:                06:40:14    BIC          708.0
05
Sample:                01-01-2022    HQIC          701.7
71
                        - 08-28-2022
Covariance Type:                opg
=====
==
                        coef    std err          z      P>|z|      [0.025      0.97
5]
-----
--
const          -0.0308      0.229      -0.135      0.893      -0.479      0.4
18
ar.L1           0.7119      0.047     15.074      0.000      0.619      0.8
05
sigma2          1.0415      0.094     11.072      0.000      0.857      1.2
26
=====
=====
Ljung-Box (L1) (Q):                0.15    Jarque-Bera (JB):
0.10
Prob(Q):                0.70    Prob(JB):
0.95
Heteroskedasticity (H):                0.60    Skew:
-0.04
Prob(H) (two-sided):                0.03    Kurtosis:
3.05
=====
=====

```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Estimated coefficient: 0.7119

True coefficient: 0.7

Measure (AR) Model Performance using MAE and RMSE

```

In [64]: # Forecast
forecast = ar_fit.forecast(steps=len(test))

# Evaluation

```

```

from sklearn.metrics import mean_absolute_error, root_mean_squared_error

rmse = root_mean_squared_error(test, forecast)
mae = mean_absolute_error(test, forecast)

print(f"Test Min: {test.min():.3f}")
print(f"Test Max: {test.max():.3f}")

print(f"AR(1) RMSE: {rmse:.3f}")
print(f"AR(1) MAE: {mae:.3f}")

```

```

Test Min: -3.629
Test Max: 3.168
AR(1) RMSE: 1.552
AR(1) MAE: 1.239

```

How do we read these results

The baseline is the actual test values. MAE and RMSE measure how far our forecasts are from the true observed values in the test set.

MAE = 1.239 means that, on average, our forecast is about 1.24 units away from the true value at each time step. RMSE = 1.552 means the typical error magnitude is about 1.55 units, with large errors penalized more.

The test values range roughly from -3.63 to 3.17, so the total amplitude is about 6.8 units. An MAE of 1.239 means that, on average, our forecast misses the true value by about 18% of the full range. An RMSE of 1.552 means typical errors are about 23% of the range, with larger mistakes weighted more heavily.

Interpretation: AR(1) captures the mean-reverting behavior but explains only a limited portion of the variability. Errors of this size indicate weak predictive power relative to the data scale. To judge usefulness, we should compare these numbers against a naive or mean forecast; if AR(1) does not reduce RMSE/MAE, it adds no forecasting value.

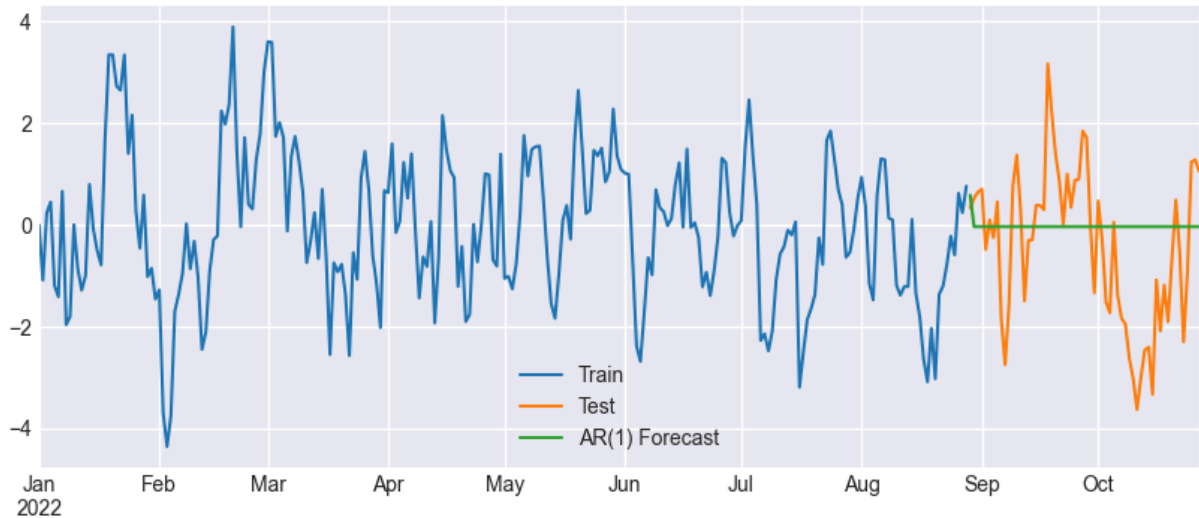
Plot the 3 series, train, test, forecast

```

In [40]: import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
train.plot(label="Train")
test.plot(label="Test")
forecast.plot(label="AR(1) Forecast")
plt.legend()
plt.show()

```



AR(1) forecast is flat because multi-step AR(1) forecasts converge to the series mean when $|\phi| < 1$. The model predicts the expected value, not future shocks, so it cannot track the volatility seen in the test data.

ϕ (phi) is the autoregressive coefficient in an AR(1) model. It measures how strongly the current value depends on the previous value.

Interpreting AR(1) Model Output - OPTIONAL

What this really is (in one sentence)

AR(1) is just linear regression where the predictor is yesterday's value.

Connect it to regular regression (intuition first)

In ordinary regression you write:

$$y = \beta_0 + \beta_1 x_1 + \text{error}$$

In AR(1), you write:

$$y_t = \text{constant} + \beta_1 \times y_{t-1} + \text{error}$$

Same idea. The "feature" is just the lagged value of y itself.

Translate the output into regression language

1. The coefficients

```
const    = -0.0308
ar.L1    =  0.7119
sigma2   =  1.0415
```

Think of this as a regression table:

Feature	Coefficient	What it means
Intercept	-0.031	Baseline (when $y_{t-1} = 0$)
y_{t-1}	0.7119	Today = 71% of yesterday + noise
Error variance	1.042	Residual spread

The prediction equation is:

$$y_t = -0.031 + 0.7119 \times y_{t-1} + \varepsilon_t$$

This is identical to:

$$\text{sales_today} = -0.031 + 0.7119 \times \text{sales_yesterday} + \text{noise}$$

2. Standard errors and p-values

Exactly the same as regression:

	coef	std err	p-value
const	-0.031	0.229	0.893
ar.L1	0.712	0.047	0.000

const: $p = 0.893$ (high)

Not significant. Could drop it.

ar.L1: $p = 0.000$ (very low)

Highly significant. Yesterday strongly predicts today.

Same interpretation as:

$$\text{feature_1: } p = 0.893 \rightarrow \text{not useful} \quad \text{feature_2: } p = 0.000 \rightarrow \text{very useful}$$

The model found **strong autocorrelation** in the data.

3. Model quality metrics

$$\begin{aligned} \text{AIC} &= 697.6 \\ \text{BIC} &= 708.0 \\ \text{Log Likelihood} &= -345.8 \end{aligned}$$

AIC stands for **Akaike Information Criterion** and BIC stands for **Bayesian Information Criterion**.

These are **goodness-of-fit measures**, not accuracy metrics.

Both measure **model quality by balancing fit and complexity**. They reward models that fit the data well and penalize models with more parameters.

AIC focuses more on predictive performance and tends to favor slightly more complex models. BIC penalizes complexity more strongly and tends to select simpler models, especially with large samples.

AIC and BIC: Lower is better

Use these to compare models:

- AR(1): AIC = 697.6
- AR(2): AIC = 695.3 → better
- MA(1): AIC = 701.2 → worse

Log Likelihood: Higher is better

Similar to minimizing loss in ML. Maximum likelihood estimation finds coefficients that make the observed data most probable.

4. Diagnostic tests (are residuals well-behaved?)

Ljung-Box (Q) :	0.15	p = 0.70
Jarque-Bera (JB) :	0.10	p = 0.95
Heteroskedasticity (H) :	0.60	p = 0.03

Think of these as **residual plots in test form**.

Ljung-Box test

Question: "Do residuals have patterns left?"

- $p > 0.05$ → No patterns (good)
- $p < 0.05$ → Patterns remain (bad)

Here: $p = 0.70$ → residuals look random

This is like checking:

 plot(residuals) shows no trend

Jarque-Bera test

Question: "Are residuals normally distributed?"

- $p > 0.05$ → Yes (good)
- $p < 0.05$ → No (may need transformation)

Here: $p = 0.95$ → very normal

This is like checking:

■ histogram(residuals) looks bell-shaped

Heteroskedasticity test

Question: "Is variance constant over time?"

- $p > 0.05 \rightarrow$ Yes (good)
- $p < 0.05 \rightarrow$ No (variance changes)

Here: $p = 0.03 \rightarrow$ some heteroskedasticity detected

This is like seeing:

■ residual spread increases over time

5. Model validation

Estimated coefficient: 0.7119
True coefficient: 0.7000

The model recovered the true data-generating process.

In ML terms:

■ The model learned the correct function

This would be like:

- You generate data: $y = 2x + \text{noise}$
- Your model learns: $y = 1.98x$
- Close match $\rightarrow$ model works

Why it feels alien

1. No feature matrix visible

The "feature" (y_{t-1}) is created internally from the time series.

2. Maximum likelihood instead of MSE

Same goal (fit the data), different math. Likelihood is more general than squared error.

3. Heavy focus on diagnostics

Econometrics cares deeply about **why** the model works, not just **that** it works.

Statistical inference > predictive accuracy

4. Different vocabulary

- ML says: " $R^2 = 0.85$ "
- Econometrics says: "Log Likelihood = -345, AIC = 697"

Same information, different packaging.

How to read this output (decision rules)

Check coefficients:

- $p < 0.05 \rightarrow$ significant $\rightarrow$ keep
- $p > 0.05 \rightarrow$ not significant $\rightarrow$ consider dropping

Check diagnostics:

- Ljung-Box $p > 0.05 \rightarrow$ good (no autocorrelation left)
- Ljung-Box $p < 0.05 \rightarrow$ bad (model incomplete)

Compare models:

- Lower AIC $\rightarrow$ better model
- Lower BIC $\rightarrow$ better model (penalizes complexity more)

Validate:

- Residuals should look random
- Residuals should be roughly normal
- Variance should be constant (ideally)

One grounding sentence you can remember

AR(1) is linear regression where X is yesterday's y, and the output tells you both fit quality and residual behavior in statistical language instead of ML metrics.

Quick interpretation of this specific output

Model found: $y_t = 0.71 \times y_{t-1} + \text{noise}$

Constant term: Not significant ($p = 0.89$) $\rightarrow$ probably zero in reality

Autocorrelation: Strong (coefficient = 0.71, $p < 0.001$) $\rightarrow$ yesterday matters a lot

Residuals: Clean (Ljung-Box $p = 0.70$, JB $p = 0.95$) $\rightarrow$ no patterns left

Minor issue: Slight heteroskedasticity ($p = 0.03$) $\rightarrow$ variance not perfectly constant

Conclusion: This is a good model. The AR(1) structure correctly captures the data-generating process.

Model 2: MA (Moving Average)

Theory

An MA(q) model predicts the current value using past forecast errors:

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_\varphi \varepsilon_{t-\varphi}$$

Where:

- μ is the mean
- $\theta_1, \theta_2, \dots, \theta_\varphi$ are coefficients
- $\varepsilon_t, \varepsilon_{t-1}, \dots$ are error terms

Implementation

The code below **fits a moving-average model of order 1 to the training data**.

More specifically:

- It fits an **ARIMA(0,0,1)** model, which is an **MA(1)** model.
- The model assumes the series depends on **one past shock (error term)** rather than past values.
- It estimates the MA coefficient using the **training data**.
- It prints a **model summary** with parameter estimates and diagnostics.

In short:

The code fits an MA(1) model so it can be compared with the AR(1) model on the same data.

```
In [38]: # Fit MA(1) model
ma_model = ARIMA(train, order=(0, 0, 1))
ma_fit = ma_model.fit()

print("MA(1) Model Summary:")
print(ma_fit.summary())
```

MA(1) Model Summary:

SARIMAX Results

```

=====
==
Dep. Variable:          Value  No. Observations:          2
40
Model:                ARIMA(0, 0, 1)  Log Likelihood          -370.4
31
Date:                Sun, 04 Jan 2026  AIC              746.8
63
Time:                06:57:15  BIC              757.3
05
Sample:                01-01-2022  HQIC              751.0
70
                        - 08-28-2022
Covariance Type:                opg
=====
==
                        coef    std err          z      P>|z|      [0.025    0.97
5]
-----
--
const          -0.0349      0.120     -0.290      0.771     -0.270      0.2
00
ma.L1           0.6287      0.051     12.246      0.000      0.528      0.7
29
sigma2          1.2801      0.119     10.766      0.000      1.047      1.5
13
=====
=====
Ljung-Box (L1) (Q):                10.99  Jarque-Bera (JB):
0.40
Prob(Q):                0.00  Prob(JB):
0.82
Heteroskedasticity (H):                0.60  Skew:
-0.10
Prob(H) (two-sided):                0.03  Kurtosis:
2.94
=====
=====

```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Measure (MA) Model Performance using MAE and RMSE

```

In [ ]: # Forecast
forecast = ma_fit.forecast(steps=len(test))

# Evaluation
from sklearn.metrics import mean_absolute_error, root_mean_squared_error

rmse = root_mean_squared_error(test, forecast)

```

```
mae = mean_absolute_error(test, forecast)

print(f"MA(1) RMSE: {rmse:.3f}")
print(f"MA(1) MAE: {mae:.3f}")
```

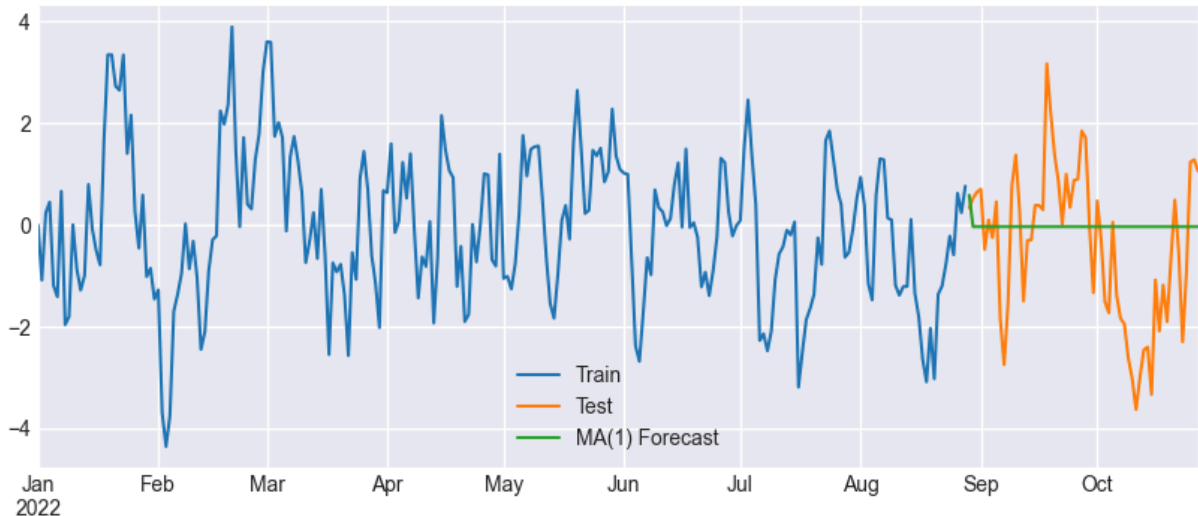
AR(1) RMSE: 1.554

AR(1) MAE: 1.252

Plot the 3 series, train, test, forecast

```
In [42]: import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
train.plot(label="Train")
test.plot(label="Test")
forecast.plot(label="MA(1) Forecast")
plt.legend()
plt.show()
```



Model 3: ARMA

Theory

ARMA(p,q) combines AR(p) and MA(q):

$$Y_t = c + \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

Important: ARMA requires stationary data.

Implementation

The code below **fits a combined autoregressive and moving-average model of order (1,1)** to the training data.

More specifically:

- It fits an **ARIMA(1,0,1)** model, also known as **ARMA(1,1)**.
- The model allows the series to depend on:
 - **one past value** (AR part), and
 - **one past shock or error** (MA part).
- The model is trained on the **training dataset**.
- It prints a **summary** showing parameter estimates and diagnostics.

In short:

The code fits an ARMA(1,1) model so its performance and parameters can be compared against the simpler AR(1) and MA(1) models.

```
In [44]: # Fit ARMA(1,1) model
arma_model = ARIMA(train, order=(1, 0, 1))
arma_fit = arma_model.fit()

print("ARMA(1,1) Model Summary:")
print(arma_fit.summary())
```

ARMA(1,1) Model Summary:

SARIMAX Results

```

=====
==
Dep. Variable:          Value    No. Observations:          2
40
Model:                ARIMA(1, 0, 1)    Log Likelihood          -345.6
71
Date:                Sun, 04 Jan 2026    AIC          699.3
41
Time:                07:00:25    BIC          713.2
64
Sample:                01-01-2022    HQIC          704.9
51
                        - 08-28-2022
Covariance Type:                opg
=====
==
                        coef    std err          z      P>|z|      [0.025      0.97
5]
-----
--
const          -0.0308      0.223     -0.138      0.890     -0.468      0.4
07
ar.L1           0.6912      0.072      9.667      0.000      0.551      0.8
31
ma.L1           0.0424      0.096      0.441      0.659     -0.146      0.2
31
sigma2          1.0405      0.094     11.083      0.000      0.857      1.2
25
=====
=====
Ljung-Box (L1) (Q):          0.00    Jarque-Bera (JB):
0.13
Prob(Q):          0.96    Prob(JB):
0.94
Heteroskedasticity (H):          0.60    Skew:
-0.05
Prob(H) (two-sided):          0.02    Kurtosis:
3.05
=====
=====

```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Measure (ARMA) Model Performance using MAE and RMSE

```

In [ ]: # Forecast
forecast = arma_fit.forecast(steps=len(test))

# Evaluation
from sklearn.metrics import mean_absolute_error, root_mean_squared_error

```

```
rmse = root_mean_squared_error(test, forecast)
mae = mean_absolute_error(test, forecast)

print(f"ARMA(1,1) RMSE: {rmse:.3f}")
print(f"ARMA(1,1) MAE: {mae:.3f}")
```

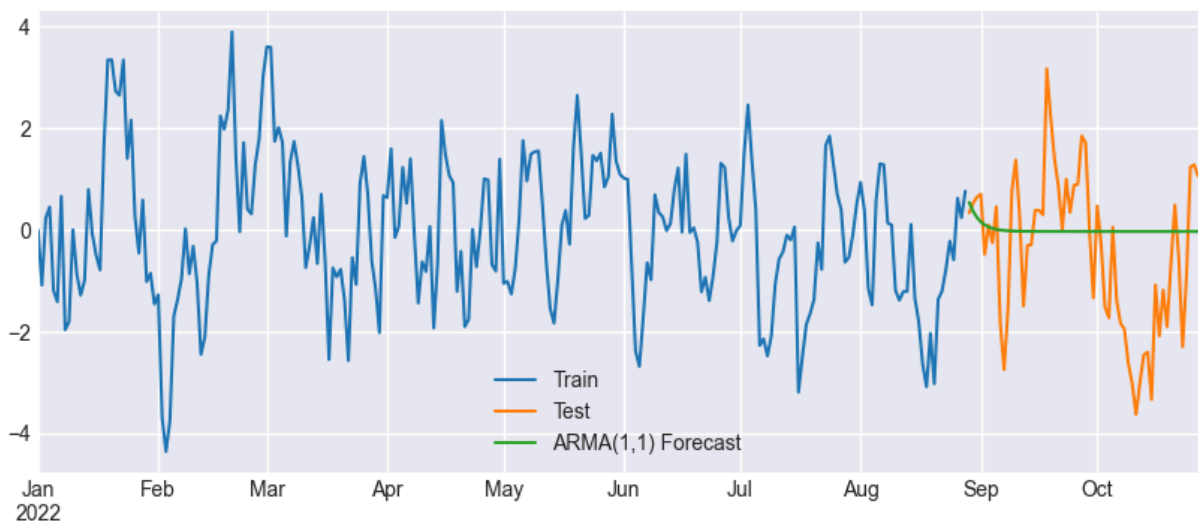
ARMA(1) RMSE: 1.552

ARMA(1) MAE: 1.239

Plot the 3 series, train, test, forecast

```
In [55]: import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
train.plot(label="Train")
test.plot(label="Test")
forecast.plot(label="ARMA(1,1) Forecast")
plt.legend()
plt.show()
```



Model 4: ARIMA

Theory

ARIMA(p,d,q) is ARMA applied to d-times differenced data:

- **p**: Order of autoregressive part
- **d**: Degree of differencing
- **q**: Order of moving average part

Steps:

1. Difference the series d times to make it stationary

2. Apply ARMA(p,q) to the differenced series

Implementation

The code below **fits a full ARIMA model, evaluates it, and uses it to make forecasts**.

More specifically:

- It fits an **ARIMA(1,1,1)** model:
 - one autoregressive term,
 - one differencing step to remove trend,
 - one moving-average term.
- The model is trained on the **training data**.
- It prints a **model summary** with estimated parameters.
- It reports **AIC and BIC**, which measure model quality while penalizing complexity (lower is better).
- It then **predicts future values** for the same number of time steps as the test set.

In short:

The code builds a trend-aware ARIMA model and uses it to forecast the unseen portion of the time series.

```
In [ ]: # Fit ARIMA(1,1,1)
        arima_model = ARIMA(train, order=(1, 1, 1))
        arima_fit = arima_model.fit()

        print("ARIMA(1,1,1) Model Summary:")
        print(arima_fit.summary())
        print(f"\nAIC: {arima_fit.aic:.2f}")
        print(f"BIC: {arima_fit.bic:.2f}")
```

ARIMA(1,1,1) Model Summary:

SARIMAX Results

=====						
==						
Dep. Variable:	Value		No. Observations:		2	
40						
Model:	ARIMA(1, 1, 1)		Log Likelihood		-346.3	
35						
Date:	Sun, 04 Jan 2026		AIC		698.6	
70						
Time:	07:01:49		BIC		709.1	
00						
Sample:	01-01-2022		HQIC		702.8	
73						
	- 08-28-2022					
Covariance Type:	opg					
=====						
==						
	coef	std err	z	P> z	[0.025	0.97
5]						

--						
ar.L1	0.7190	0.052	13.919	0.000	0.618	0.8
20						
ma.L1	-0.9999	2.479	-0.403	0.687	-5.860	3.8
60						
sigma2	1.0460	2.583	0.405	0.685	-4.016	6.1
08						
=====						
=====						
Ljung-Box (L1) (Q):	0.08		Jarque-Bera (JB):			
0.08						
Prob(Q):	0.78		Prob(JB):			
0.96						
Heteroskedasticity (H):	0.59		Skew:			
-0.03						
Prob(H) (two-sided):	0.02		Kurtosis:			
3.06						
=====						
=====						

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

AIC: 698.67

BIC: 709.10

Measure (ARIMA) Model Performance using MAE and RMSE

```
In [ ]: # Forecast
forecast = arima_fit.forecast(steps=len(test))
```

```
# Evaluation
from sklearn.metrics import mean_absolute_error, root_mean_squared_error

rmse = root_mean_squared_error(test, forecast)
mae = mean_absolute_error(test, forecast)

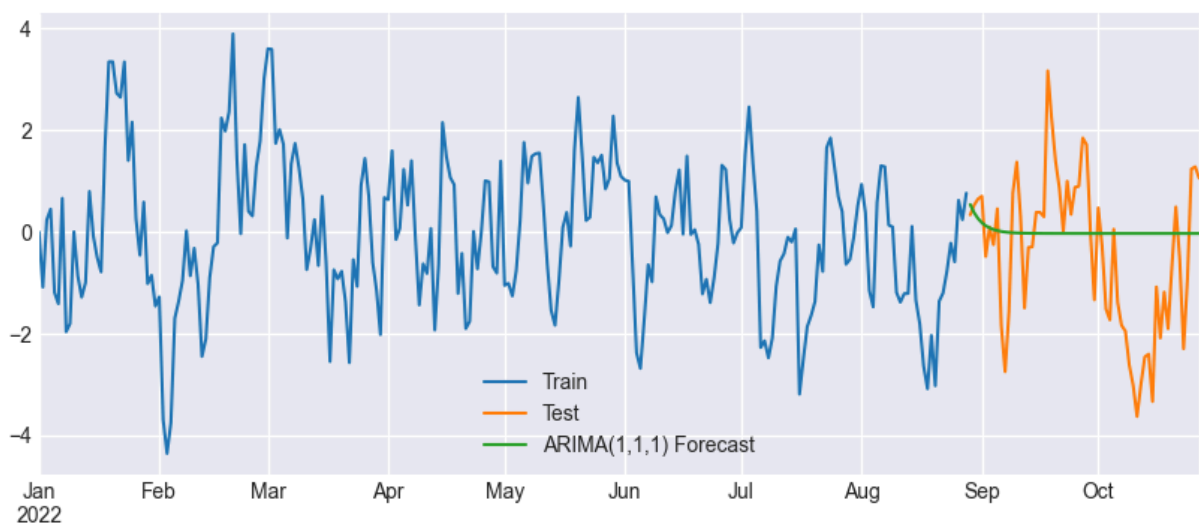
print(f"ARIMA(1,1,1) RMSE: {rmse:.3f}")
print(f"ARIMA(1,1,1) MAE: {mae:.3f}")
```

AR(1) RMSE: 1.552
AR(1) MAE: 1.239

Plot the 3 series, train, test, forecast

```
In [53]: import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
train.plot(label="Train")
test.plot(label="Test")
forecast.plot(label="ARIMA(1,1,1) Forecast")
plt.legend()
plt.show()
```



Model 5: SARIMA (Seasonal ARIMA)

Theory

SARIMA(p,d,q)(P,D,Q,s) extends ARIMA to handle seasonality:

Non-seasonal part: (p,d,q) **Seasonal part:** (P,D,Q,s)

- P: Seasonal AR order
- D: Seasonal differencing order
- Q: Seasonal MA order

- s: Seasonal period (e.g., 12 for monthly data with yearly seasonality)

Creating Seasonal Data

The code below **creates fake monthly data that looks like long-term seasonal sales and splits it into training and test sets**.

More specifically:

- It creates **20 years of monthly dates**.
- It builds data with:
 - a **slow upward trend** over time,
 - a **repeating yearly pattern** (seasonality every 12 months),
 - and some **random noise**.
- These components are added together to look like realistic sales data.
- The data is then split into:
 - **80% for training**,
 - **20% for testing**.
- Finally, it prints the size of the full series and each split.

In short:

The code simulates a realistic monthly time series with trend and seasonality and prepares it for seasonal time-series modeling.

```
In [66]: # Generate monthly data with seasonality
n_months = 240 # 20 years
dates_monthly = pd.date_range(start='2004-01-01', periods=n_months, freq='MS')

# Components
trend = np.linspace(50, 100, n_months)
seasonal = 15 * np.sin(2 * np.pi * np.arange(n_months) / 12)
noise = np.random.normal(0, 3, n_months)

ts_seasonal = pd.Series(trend + seasonal + noise,
                        index=dates_monthly,
                        name='Sales')

# Split
train_seas = ts_seasonal[:int(len(ts_seasonal) * 0.8)]
test_seas = ts_seasonal[int(len(ts_seasonal) * 0.8):]

print(f"Seasonal series shape: {ts_seasonal.shape}")
print(f"Train: {len(train_seas)}, Test: {len(test_seas)}")
```

Seasonal series shape: (240,)
Train: 192, Test: 48

Fitting SARIMA

The code below **fits a seasonal ARIMA model and uses it to make forecasts**.

More specifically:

- It fits a **SARIMA(1,1,1)(1,1,1,12)** model, which means:
 - one autoregressive term, one differencing, and one moving-average term for the **non-seasonal part**,
 - one autoregressive term, one differencing, and one moving-average term for the **seasonal part**,
 - with a **12-month seasonal cycle**.
- The model is trained on the **seasonal training data**.
- It prints a **summary** showing estimated parameters and diagnostics.
- It then **forecasts future values** for the length of the test period.

In short:

The code models both long-term trends and repeating yearly patterns, then uses that model to predict future seasonal behavior.

```
In [67]: # Fit SARIMA(1,1,1)(1,1,1,12)
# Seasonal period = 12 months
sarima_model = SARIMAX(train_seas,
                        order=(1, 1, 1),
                        seasonal_order=(1, 1, 1, 12))
sarima_fit = sarima_model.fit(dispatch=False)

print("SARIMA(1,1,1)(1,1,1,12) Model Summary:")
print(sarima_fit.summary())

# Forecast
sarima_forecast = sarima_fit.forecast(steps=len(test_seas))
```

SARIMA(1,1,1)(1,1,1,12) Model Summary:

SARIMAX Results

=====						
Dep. Variable:		Sales		No. Observations:		
192						
Model:		SARIMAX(1, 1, 1)x(1, 1, 1, 12)		Log Likelihood		
-463.423						
Date:		Sun, 04 Jan 2026		AIC		
936.845						
Time:		07:27:04		BIC		
952.782						
Sample:		01-01-2004		HQIC		
943.307						
		- 12-01-2019				
Covariance Type:		opg				
=====						
==						
	coef	std err	z	P> z	[0.025	0.97
5]						

--						
ar.L1	-0.0128	0.084	-0.151	0.880	-0.178	0.1
52						
ma.L1	-0.9801	0.040	-24.634	0.000	-1.058	-0.9
02						
ar.S.L12	-0.0238	0.104	-0.230	0.818	-0.227	0.1
79						
ma.S.L12	-0.9156	0.140	-6.530	0.000	-1.190	-0.6
41						
sigma2	8.9146	1.336	6.671	0.000	6.295	11.5
34						
=====						
=====						
Ljung-Box (L1) (Q):		0.03	Jarque-Bera (JB):			
1.26						
Prob(Q):		0.87	Prob(JB):			
0.53						
Heteroskedasticity (H):		1.30	Skew:			
-0.07						
Prob(H) (two-sided):		0.31	Kurtosis:			
2.61						
=====						
=====						

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Measure (SARIMA) Model Performance using MAE and RMSE

```
In [68]: # Forecast
forecast = sarima_fit.forecast(steps=len(test_seas))
```

```
# Evaluation
from sklearn.metrics import mean_absolute_error, root_mean_squared_error

rmse = root_mean_squared_error(test_seas, forecast)
mae = mean_absolute_error(test_seas, forecast)

print(f"SARIMA(1,1,1)(1,1,1,12) RMSE: {rmse:.3f}")
print(f"SARIMA(1,1,1)(1,1,1,12) MAE: {mae:.3f}")
```

SARIMA(1,1,1)(1,1,1,12) RMSE: 2.703

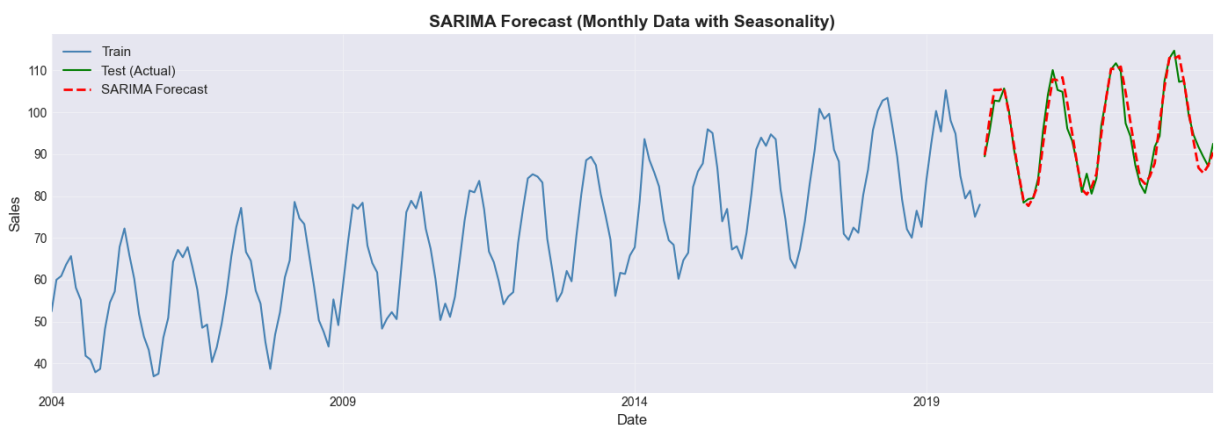
SARIMA(1,1,1)(1,1,1,12) MAE: 2.056

Plot the 3 series, train, test, forecast

```
In [69]: fig, ax = plt.subplots(figsize=(14, 5))

train_seas.plot(ax=ax, label='Train', linewidth=1.5, color='steelblue')
test_seas.plot(ax=ax, label='Test (Actual)', linewidth=1.5, color='green')
sarima_forecast.plot(ax=ax, label='SARIMA Forecast', linewidth=2,
                    linestyle='--', color='red')

ax.set_title('SARIMA Forecast (Monthly Data with Seasonality)',
            fontsize=14, fontweight='bold')
ax.set_xlabel('Date', fontsize=12)
ax.set_ylabel('Sales', fontsize=12)
ax.legend(fontsize=11)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



All Content Below is OPTIONAL

6. Model Selection

Information Criteria

AIC (Akaike Information Criterion)

$$\text{AIC} = 2k - 2\ln(L)$$

Where:

- k = number of parameters
- L = maximum likelihood

BIC (Bayesian Information Criterion)

$$\text{BIC} = k \cdot \ln(n) - 2\ln(L)$$

Where:

- n = number of observations

Rule: Lower AIC/BIC = better model (balances fit and complexity)

Grid Search for Best Model

```
In [ ]: def evaluate_arma_models(data, p_range, d_range, q_range):
    """
    Evaluate different ARIMA configurations
    """
    results = []

    for p in p_range:
        for d in d_range:
            for q in q_range:
                try:
                    model = ARIMA(data, order=(p, d, q))
                    fitted = model.fit()

                    results.append({
                        'order': (p, d, q),
                        'AIC': fitted.aic,
                        'BIC': fitted.bic
                    })
                except:
                    continue

    return pd.DataFrame(results)

# Search for best model
print("Searching for best ARIMA model...")
results_df = evaluate_arma_models(
```



```

train,
p_range=range(0, 3),
d_range=range(0, 2),
q_range=range(0, 3)
)

# Sort and display
results_df = results_df.sort_values('AIC')
print("\nTop 5 Models by AIC:")
print(results_df.head())

best_order = results_df.iloc[0]['order']
print(f"\nBest Model: ARIMA{best_order}")

```

Searching for best ARIMA model...

Top 5 Models by AIC:

	order	AIC	BIC
6	(1, 0, 0)	697.563191	708.005108
13	(2, 0, 1)	698.028252	715.431446
10	(1, 1, 1)	698.670168	709.099558
12	(2, 0, 0)	699.335230	713.257786
7	(1, 0, 1)	699.341125	713.263681

Best Model: ARIMA(1, 0, 0)

7. Model Diagnostics

Why Check Diagnostics?

After fitting a model, we need to verify that:

1. **Residuals are white noise** (random, no pattern)
2. **Residuals are normally distributed**
3. **No autocorrelation in residuals**
4. **Model assumptions are satisfied**

Theory: Good Residuals

If the model is appropriate, residuals should have:

- Mean ≈ 0
- Constant variance (homoscedastic)
- No autocorrelation
- Normal distribution

Diagnostic Plots

```
In [ ]: # Get residuals from best model
best_model = ARIMA(train, order=best_order).fit()
residuals = best_model.resid

# Create diagnostic plots
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

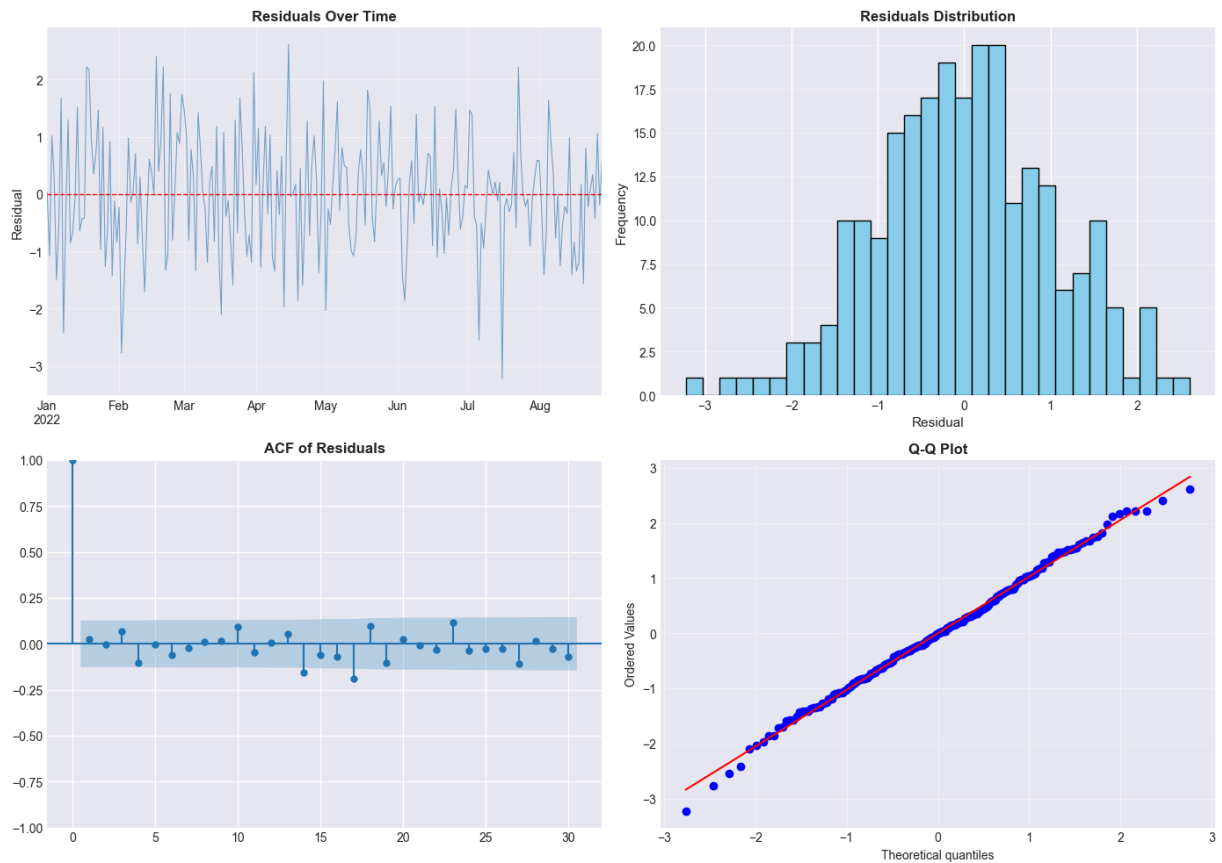
# 1. Residuals over time
residuals.plot(ax=axes[0, 0], linewidth=0.8, color='steelblue', alpha=0.7)
axes[0, 0].axhline(y=0, color='red', linestyle='--', linewidth=1)
axes[0, 0].set_title('Residuals Over Time', fontsize=12, fontweight='bold')
axes[0, 0].set_ylabel('Residual', fontsize=11)
axes[0, 0].grid(True, alpha=0.3)

# 2. Histogram
residuals.hist(ax=axes[0, 1], bins=30, edgecolor='black', color='skyblue')
axes[0, 1].set_title('Residuals Distribution', fontsize=12, fontweight='bold')
axes[0, 1].set_xlabel('Residual', fontsize=11)
axes[0, 1].set_ylabel('Frequency', fontsize=11)
axes[0, 1].grid(True, alpha=0.3, axis='y')

# 3. ACF of residuals
plot_acf(residuals, lags=30, ax=axes[1, 0])
axes[1, 0].set_title('ACF of Residuals', fontsize=12, fontweight='bold')

# 4. Q-Q plot
stats.probplot(residuals, dist="norm", plot=axes[1, 1])
axes[1, 1].set_title('Q-Q Plot', fontsize=12, fontweight='bold')
axes[1, 1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



Ljung-Box Test

Tests the null hypothesis that residuals are independently distributed (no autocorrelation).

- H_0 : Residuals are white noise (no autocorrelation)
- H_1 : Residuals have autocorrelation

Decision: p-value > 0.05 → residuals are white noise (good!)

```
In [ ]: # Perform Ljung-Box test
lb_test = acorr_ljungbox(residuals, lags=[10, 20, 30], return_df=True)

print("\nLjung-Box Test Results:")
print(lb_test)
print("\nInterpretation:")
print("If p-value > 0.05: Residuals are white noise ✓")
print("If p-value < 0.05: Residuals have autocorrelation ✗")
```

```
Ljung-Box Test Results:
      lb_stat  lb_pvalue
10    7.147022   0.711496
20   31.838671   0.045048
30   41.570079   0.077860
```

Interpretation:

If p-value > 0.05: Residuals are white noise ✓

If p-value < 0.05: Residuals have autocorrelation ✗

Residual Statistics

```
In [ ]: print("\nResidual Summary Statistics:")
print("="*50)
print(f"Mean:      {residuals.mean():>10.6f}  (should be ≈ 0)")
print(f"Std Dev:   {residuals.std():>10.6f}")
print(f"Min:       {residuals.min():>10.6f}")
print(f"Max:       {residuals.max():>10.6f}")
print(f"Skewness:   {residuals.skew():>10.6f}  (should be ≈ 0)")
print(f"Kurtosis:   {residuals.kurtosis():>10.6f}  (should be ≈ 0)")
```

Residual Summary Statistics:

```
=====
Mean:      -0.000090  (should be ≈ 0)
Std Dev:   1.022687
Min:      -3.222888
Max:       2.614799
Skewness:  -0.045184  (should be ≈ 0)
Kurtosis:   0.071627  (should be ≈ 0)
```

8. Model Evaluation

Common Forecasting Metrics

1. MAE (Mean Absolute Error)

$$\text{MAE} = (1/n) \sum |y_i - \hat{y}_i|$$

- Average absolute difference
- Same units as original data
- Easy to interpret

2. RMSE (Root Mean Squared Error)

$$\text{RMSE} = \sqrt{(1/n) \sum (y_i - \hat{y}_i)^2}$$

- Penalizes large errors more
- Same units as original data
- More sensitive to outliers than MAE

3. MAPE (Mean Absolute Percentage Error)

$$\text{MAPE} = (100/n) \sum |(y_i - \hat{y}_i)/y_i|$$

- Expressed as percentage
- Scale-independent

- Undefined when $y_i = 0$

Computing Metrics

```
In [ ]: def calculate_metrics(actual, predicted):
        """Calculate forecasting metrics"""
        mae = mean_absolute_error(actual, predicted)
        rmse = np.sqrt(mean_squared_error(actual, predicted))
        mape = np.mean(np.abs((actual - predicted) / actual)) * 100

        return {'MAE': mae, 'RMSE': rmse, 'MAPE': mape}

# Evaluate forecast
forecast = best_model.forecast(steps=len(test))
metrics = calculate_metrics(test, forecast)

print("\nForecast Evaluation Metrics:")
print("="*50)
print(f"MAE: {metrics['MAE']:.4f}")
print(f"RMSE: {metrics['RMSE']:.4f}")
print(f"MAPE: {metrics['MAPE']:.2f}%")
```

Forecast Evaluation Metrics:

```
=====
MAE: 1.2387
RMSE: 1.5522
MAPE: 102.35%
```

Summary and Key Takeaways

Workflow for Time Series Modeling

1. Explore Data

- Plot the series
- Identify trend, seasonality, noise
- Perform decomposition

2. Check Stationarity

- Use ADF test
- Apply differencing if needed
- Verify stationarity after transformation

3. Identify Model Orders

- Examine ACF plot → suggests MA order (q)
- Examine PACF plot → suggests AR order (p)
- Consider seasonal patterns

4. Fit Candidate Models

- Start with simple models (AR, MA)
- Try ARIMA for non-stationary data
- Use SARIMA for seasonal data

5. Select Best Model

- Compare AIC/BIC values
- Lower is better
- Balance complexity and fit

6. Validate Model

- Check residual plots
- Perform Ljung-Box test
- Ensure residuals are white noise

7. Forecast and Evaluate

- Generate predictions
- Calculate MAE, RMSE, MAPE
- Compare with test data

Model Quick Reference

If you see...	Consider...
Clear trend	Differencing ($d > 0$) or ARIMA
Seasonality	SARIMA with appropriate period
PACF cuts off at lag p	AR(p)
ACF cuts off at lag q	MA(q)
Both decay slowly	ARMA or ARIMA
Non-constant variance	Log transformation or multiplicative model

Python Libraries

- **pandas**: Data manipulation and time series structures
- **statsmodels**: ARIMA, SARIMA, statistical tests, diagnostics
- **matplotlib**: Visualization
- **scipy**: Statistical functions
- **sklearn**: Evaluation metrics

Next Steps

In **Part 2**, we'll cover:

- Advanced pandas techniques for time series
- Resampling and frequency conversion
- Rolling windows and moving averages
- Time-based indexing and slicing
- Practical data preparation workflows

In **Part 3**, we'll explore:

- Feature engineering for ML models
- Machine learning approaches (XGBoost, Random Forest)
- Deep learning (LSTM, GRU)
- Prophet and modern forecasting tools

End of Part 1